

EDUCATION

- **University of Illinois at Urbana Champaign (UIUC)** *Aug. 2019 – Present*
Ph.D. in Computer Science *Champaign, IL*
- **National Taiwan University** *Feb. 2015 – Sept. 2018*
M.S. in Computer Science, GPA: 4.13/4.30 *Taipei, Taiwan*

SKILLS AND INTERESTS

Machine Learning Python, Pytorch, CNN, Transformer, Large language model (LLM)
Computer Vision Continual learning, 360° video content analysis, Vision language model (VLM)

WORK EXPERIENCE

- **Alexa @ Amazon** *July – Nov. 2022*
Applied Scientist Intern *Champaign, IL (Remote)*
 - Enhanced Alexa's directed voice activity detection performance by ~15% with multimodal fusion of video and audio, improving user interaction quality.
 - Explored customized fine-tuning and further decreased false reject rate by over 5%.
- **VISA Research** *May – July 2020; May – Aug. 2021*
Machine Learning Researcher Intern *Champaign, IL (Remote)*
 - Investigated settlement volume prediction with time series forecasting models.
 - Pioneered a reinforcement learning model for foreign exchange rate prediction, achieving a 53% accuracy in trend forecasting.

RESEARCH EXPERIENCE

- **UIUC (w/ Prof. Klara Nahrstedt & Yuxiong Wang)** *Aug. 2019 – Present*
 - Discovered critical limitations in **continual learning** algorithm development with from-scratch training.
 - Reduced computation and client device requirements in **360° video streaming** via offline field of view prediction.
- **Research assistant to Prof. Winston Hsu** *Feb. 2015 – July 2019*
 - Reduced mean absolute error in solar irradiance prediction by 21%, leveraging **satellite images** and Sun angles.
 - Advanced **video inpainting** by reducing the FID over 20% compared to the state-of-the-art models.

PROFESSIONAL SERVICE

- *Reviewer* for Machine Learning and Computer Vision Conferences
 - CVPR'25, ICLR'25, AAAI'25, CVPR'24, ECCV'24, CVPR'23, ICLR'23, NeurIPS'22, CVPR'22, ICCV'21
- *Teaching Assistant* for Computational Photography
 - FA'20, SP'21, FA'21, SP'22, SP'23 with Prof. Derek Hoiem and Yuxiong Wang

SELECTED PUBLICATION

- [1] **K.Y. Lee**, et al. "Pseudo Dataset Generation for Out-of-domain Multi-Camera View Recommendation." *VCIP 2024* [PDF] [Project]
- [2] **K.Y. Lee**, et al. "Do Pre-Trained Models Benefit Equally in Continual Learning?." *WACV 2023* [PDF] [Project]
- [3] **K.Y. Lee**, et al. "Redefine the A in ABR for 360-degree Videos: A Flexible ABR Framework." *ISM 2020* [PDF]
- [4] **K.Y. Lee**, et al. "Metadata-Augmented Neural Networks for Cross-Location Solar Irradiation Prediction from Satellite Images." *KDD 2019* [PDF]
- [5] Y.L. Chang, Z.Y. Liu, **K.Y. Lee**, Winston Hsu. "Free-form Video Inpainting with 3D Gated Convolution and Temporal PatchGAN." *ICCV 2019* [PDF]